Extension Inspector AI

Official Product Whitepaper & Deployment Guide

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1. Executive Summary & Vision

Extension Inspector AI is the world smartest, privacy-first Browser Intelligence Platform.
Designed as a lightweight Manifest V3 Chrome extension, it explains what installed extensions do,
how they affect CPU/RAM performance, whether they can be trusted, and recommends actions in plain English.

2. Why Extension Inspector AI is Unique Over Traditional Extension Managers

â¢ Event-Driven Architecture: 0% idle CPU usage. Background service worker sleeps when inactive.
â¢ No Web Page Script Pollution: Zero React or heavy script injection into visited websites.
â¢ Flagship Extension DNA Engine: Analyzes browser ecosystem as a cohesive system rather than isolated tools.
â¢ Duplicate Feature Detector: Identifies functional redundancy (e.g. 3 PDF viewers) & calculates RAM saved.
â¢ Category Conflict Detector: Pinpoints overlapping network hooks (e.g. multiple ad blockers or password managers).
â¢ Plain-English Permission Translation: Converts raw Chrome permissions (<all_urls>, debugger) to human advice.
â¢ 100% Local & Privacy First: No remote servers, telemetry, or external data tracking.

3. Core Engine Architecture

â¢ Discovery Engine: Parses manifest specifications, permissions, icons, and host access levels.
â¢ Trust Engine: Calculates explainable Trust Scores (0-100) & Grades (A+ to F) based on deterministic criteria.
â¢ Behavior Engine: Infers script injection, network interception, cookie reading, and category types.
â¢ Privacy Engine: Evaluates host access reach (<all_urls>) and sensitive cookie exposure.
â¢ Performance Engine: Computes estimated RAM memory footprint (MB) and startup delay impact (ms).
â¢ Recommendation Engine: Provides prioritized 1-click advice (Disable, Uninstall, Review, Keep).

4. Quick Start & Usage Instructions

1. Open Google Chrome and navigate to chrome://extensions.
2. Enable Developer mode in the top-right header.
3. Click "Load unpacked" and select the extension build dist folder.
4. Click the Extension Inspector AI toolbar icon to launch the Browser Intelligence Dashboard.